

Stability of solutions of Dirichlet problems for semilinear equations with Lévy operators

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We consider sequences of solutions of semilinear Dirichlet problems with (possibly nonsymmetric) Lévy operators. We will present general results showing that if symbols of these operators converge pointwise to a symbol of some Lévy operator, then under suitable assumptions on the nonlinearity and the boundary condition these solutions converge. Examples include equations with fractional Laplace operators, relativistic fractional operators, mixed local and nonlocal operators, logarithmic Schrödinger operators, anisotropic operators, nonsymmetric stable operators. Our proofs are based on limit theorems for backward stochastic differential equations naturally associated with the equations under consideration.

My talk is based on joint work with Tomasz Klimsiak and Leszek Słomiński.